Semantic taxonomy enrichment to improve business text classification for dynamic environments

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***Abstract*—Taxonomies are widely used by various business organizations for document classification and organization. Business models built using taxonomies have the potential to reduce their efficiency with the arrival of new business ideas and concepts in the market over time. This happens because outdated business taxonomies get insufficient to fully capture the intended meaning of business documents to perform classification. Experts may need time to understand and engineering to place new data into the taxonomy to increase the classification accuracy of business documents. Here, the idea of automatic semantic enrichment of taxonomies came into consideration. This study introduces a taxonomy enrichment approach based on Natural Language Processing (NLP) technique, i.e. BERTopic, a Neural topic modeling using contextualized Bidirectional Encoder Representations from Transformers (BERT) to automatically augment a given business taxonomy with many additional concepts by leveraging a corpus of online news documents. The experiments show that augmenting topics from the text corpus into the taxonomy substantially increases the classification accuracy of business documents.**

Keywords—Business, document classification, NLP, topic modeling, transformers

I. INTRODUCTION

The term "business" is an economic activity concerning many subjects linked to; marketing, finance, risk, managerial activities, etc. which comprise a high level of decision making that needs to be supported by organized information [1 - 4]. As an important step of organizing business information, classification of the business news plays an important role in structuring the crucial market information [2]. In companies, news articles as one of the major information sources are regularly classified based on business users' requirements [1, 2]. The news articles are tagged with a finite set of keywords (concepts) from a defined business-specific taxonomy [1].

A taxonomy is a tree-based structure where concepts are organized into a hierarchy with hypernymy relationship and are used for classifying information. The taxonomies are usually constructed manually [5]. However, there are techniques present in the literature to construct business taxonomies automatically [5 - 7]. Once taxonomies are constructed, a large amount of time and effort is required to enrich them with new domain concepts. The process of adding new contextually-similar concepts to taxonomy by respecting the current hierarchy of a taxonomy is called semantic enrichment [6, 7]. The process of semantic enrichment of old taxonomies allows business users to refine their search queries. To improve the classification of the information resulting from the dynamic (i.e. changing) world news presents two-fold challenges related to the process of the semantic enrichment of taxonomy, which are; 1) identification of business-related data sources to extend taxonomies, and 2) adding new concepts to the existing taxonomy concerning the current arrangement of business concepts in a taxonomy [5 - 7].

There are various data sources that can be used to enrich taxonomy with additional concepts. Major data sources found in the existing literature [8, 9] which are used widely by the NLP community are lexical taxonomies such as WordNet (i.e. a network of synonym sets representing lexical concepts) and Wiktionary (i.e. a web-based open project with terms having words, phrases, etc.). The up-gradation of lexical taxonomies such as WordNet mainly involved the manual addition of new concepts [8]. However, the manual enrichment of WordNet is expensive, time-consuming, and needs domain experts. The 2nd most important method to enrich taxonomy that has been used in the literature are non-contextualized FastText [10] and contextualized BERT embeddings [11, 12]. The mapping of words of a document to vectors of real numbers is called embeddings [11]. Word embeddings are the word representations that allow words of a document with similar meaning to carry a similar representation. Pre-trained FastText embedding models are easy to use and do not require training data for unknown words. Whereas BERT requires a context for a word (i.e. a sentence containing it) to generate its embedding [12].

Up to now, the existing literature has a shortage of comprehensive studies which attempt to exploit state-of-the-art NLP techniques to improve business news classification using up-to-date taxonomy by capturing new market concepts in this specific domain of the business industry. In an effort to cover this gap, this article focuses on developing a taxonomy enrichment method for business news classification using existing NLP techniques. The idea of semantic enrichment of a business taxonomy for this article is constructed based on the exploitation of topic modeling using BERT embeddings [12, 13]. The BERT allows us to train the model using our latest news articles dataset to find unknown emerging topics from the changing business world and to add them to the existing taxonomy of a company.

This paper is organized as follows. Section 2 reviews the background on taxonomy enrichment and document classification. Section 3 introduces the proposed method. The method evaluation approach is described in Section 4. The discussion and the conclusion is described Section 5.

II. BACKGROUND

Topic models are proven to be a powerful tool to uncover underlying narratives (i.e. topics) in the form of keywords in text [14]. Traditional approaches, such as Latent Dirichlet Allocation (LDA) and Non-Negative Matrix Factorization (NMF) describe a text document as a bag of words and model each document as a collection of topics [13, 15]. A bag-of-words is a representation of text for describing the presence of

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words within a document [13]. It involves two elements; which are; 1) the vocabulary of words, and the frequency of the presence of words. It is known as a bag of words as it does not hold any information about the structure or order of words in the document. One of the major limitations of topic models such as LDA i.e. based on bag-of-words representation is that the semantic relationships among words are not regarded through such representations [13]. The bag-of-words representations do not consider the underlying context of words in sentences and therefore fail to accurately represent documents [13].

To address this limitation, text embedding methods [13] have quickly got popular in the NLP community. More particularly, BERT [12] and its variations [16 – 19], have demonstrated promising results in generating contextual word and sentence vector representations by considering the underlying semantic context in text documents. BERT-based embeddings allow encoding of the semantic properties of words through vector representations for arranging similar texts close in vector space [13]. In BERTopic (i.e. advanced technique after conventional LDA modeling which captures the semantic information in text), the text documents are embedded for creating representations in vector space that can be distinguished semantically [13]. The documents comprising the same topic are assumed to be semantically similar. BERTopic utilizes the Sentence-BERT (SBERT) framework [19] for executing the embedding step. This framework permits users to transform sentences into dense vector representations using language models. The vector representations are clustered based on the similarity, and finally, topic representations are mined using a custom class-based Term Frequency–Inverse Document Frequency (TF-IDF). More details on BERTopic can be found at [13].

Although BERT-based embeddings built on contextual representations for topic modeling [13] have been used widely for a variety of different NLP-related tasks, e.g. developing neural search engines, etc. However, the application of such models for taxonomy enrichment is not yet adequately explored in the literature. This study aims to discover the exploitation of the topic modeling technique based on BERTopic to extend the current business taxonomy. Using this technique, a set of relevant clusters of domain-related concepts are extracted from the corpus of news articles and added to the taxonomy. Later, to access the quality of the news articles classification using the updated taxonomy, the Count Vectorizer (CV) and the TF–IDF are employed for feature extraction [20].

III. Proposed Method

To enrich manually curated existing business taxonomy with new information from online resources such as the news articles to better model the dynamic business environment, we propose the application of state-of-the-art BERTopic [13] to enrich business taxonomy with new concepts. BERTopic is a topic model that uses clustering techniques and a class-based variation of TF-IDF (i.e. c-TF-IDF) to generate coherent topic representations. More specifically, initially, document embeddings using a pretrained language model are created to obtain document-level information [13]. Second, the dimensionality of document embeddings is reduced before generating semantically similar clusters of documents that each represents a distinct topic [13]. Lastly, a class-based version of TF-IDF is used for extracting the topic representation from each topic [13].

A. Dataset Description

To use BERTopic for creating topics to enrich the existing taxonomy, a dataset of news articles (see Fig. 1) is used. These news articles were crawled as text files from 95 different news sources through their original websites, such as lamontagne.fr, fusacq.com, businessimmo.com, zonebourse.com, francetvinfo.fr, sudouest.fr, actu.fr, ladepeche.fr, lejournaldesentreprises.com, lavoixdunord.fr, etc. The releasing time of these news articles spans five years from 2017 to 2021 (see Table 1).

TABLE I. Dataset Details

Year	2017	2018	2019	2020	2021	Total
No. of news articles	4,689	6,531	5,475	4,708	5,106	26,509

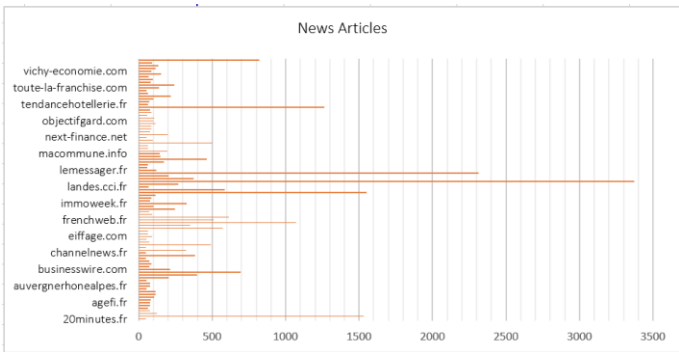

Fig. 1. Number of extracted articles from online news sources

To obtain the unique topics from online news articles, a corpus of 26,509 text files carrying 13,264,508 words is used. The languages of these text files were mainly French and English. The French text files were translated into English and basic pre-processing steps [21] such as stop-word removal, and filtering of special characters, hashtags, weblinks, numbers, email addresses, and punctuations were executed. Then, lemmatization is performed to lemmatize the words. To end, a vocabulary of 8,128,981 (concept-level term candidates) after the pre-processing and stop-word removal from the news articles is obtained.

Fig. 2. Most frequent words from online news articles

B. Applying Topic Modeling

The 1st step is to convert the preprocessed news articles to numerical data. BERT is used to extract different embeddings based on the word context [13]. For this, a python package i.e. “sentence-transformers” is exploited to capture document-level embeddings by transforming the text documents into 512-dimensional vectors [13]. In the literature, they exist many data clustering algorithms. HDBSCAN, a density-based algorithm, is chosen and applied for clustering the similar documents of the corpus [22]. It is selected because

HDBSCAN does not force less relevant data points to clusters because it takes them as outliers [13, 22]. In Fig. 3, the clustering of similar documents is achieved, which should signify the topics that they cover.

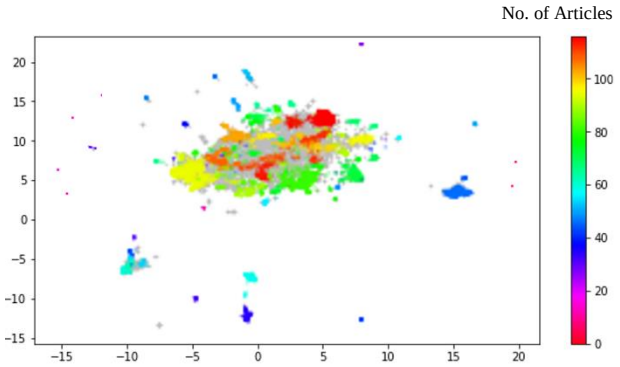

Fig. 3. Clustering similar documents together representing the topics that they consist of and plotting the outliers as grey points

After visualizing the similar documents, the BERTopic model is instatiated using a multilingual language model, trained on the corpus of news articles and topics are extracted. To understand the hierarchical organization of the topics, a method i.e. "cluster.hierarchy" from spicy Python library [13] is used to visualize how extracted topics are related to each other (see Fig. 4). The visualization will help to reduce the number of topics if required in future.

Each cluster contains a group of topics. These topics are visualized by constructing bar charts out of the c-TF-IDF scores for each topic representation. This will help to compare topic representations to each other (see Fig. 5). With the generated topic embeddings, a similarity matrix by applying cosine similarities through these embeddings is created (Fig. 6 showing the result of a similarity matrix indicating similar topics are to each other). By this visualization, similar topics can be grouped together by reducing the overall number of topics. During the insertion of new concepts into a business taxonomy, the identified new concepts, as well as taxonomy concepts, are converted into vector representations, and later similarity is used to identify semantic similarities between them. Based on their similarity score threshold, the concepts are filtered for insertion into the business taxonomy based on the cosine similarity. (see Table 2).

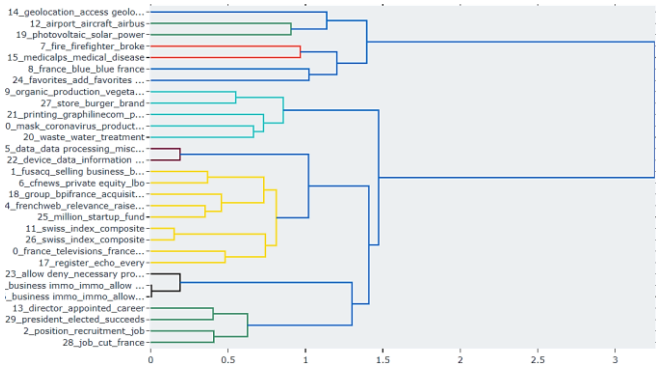

Fig. 4. Hierarchical grouping of identified clusters having topics

IV. Evaluation

After the semantic taxonomy enrichment using BERTopic modeling, we need to evaluate the effectiveness of the enriched taxonomy for news articles classification. For the taxonomy evaluation through the classification of 5,000 news articles in our case, we need 1) to transform news articles to numerical data as machines do not understand textual data, and 2) to perform a keyword identification and extraction process based on their frequency. The extracted keywords (features) will later be semantically mapped with the taxonomy concepts to perform text classification. Numerous approaches for keyword identification and extraction exist in the literature [2] at low computational costs. The two most fundamental and widely used state-of-the-art approaches are the CV and the TF–IDF vectorizer [2].

The CV provides a simple method to both tokenize (i.e. assigning a unique number to each word) text data and build a vocabulary of keywords based on their frequency. The keyword count from textual documents is very fundamental and sometimes has not proved to be efficient for text classification because of the large word count of less important words which are meaningless for the analysis. Whereas, the TF–IDF vectorizer can be effectively used for the filtering of such words.

Fig. 5. Selected terms for topics by creating bar charts out of the c-TF-IDF scores for each topic representation

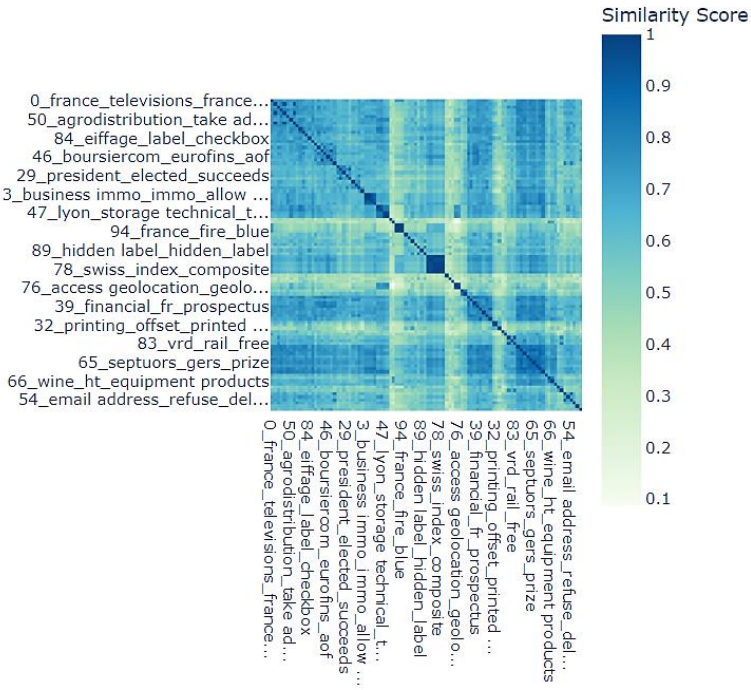

Fig. 6. Selected terms against topics by creating bar charts out of the c-TF-IDF scores for each topic representation

TABLE II. LIST OF EXTRACTED TOPICS MAPPED WITH BUSINESS TAXONOMY

No.	Taxonomy Concept	Data Enrichment Using Topics
1	arts and hobbies	sports reserved, sports, culture leisure, young, leisure, platform
2	electrical cabinet	photovoltaic, solar, power, power plant, energy, plant, electricity, panel, solar power
3	health insurance	medicalps, disease, medical, drug, treatment, patient, clinical, cancer, health, research
4	public administration	appointed, director, career, general, manager, management, joined, deputy, group, board
5	foreign affairs	european union, beneficiary, supported european, regional development, development fund, action, european regional
6	antibiotic	clinical, biotech, argobio, patient, diseases, pharma, surgivisio
7	electrodiagnostic device	battery, electric, battery factory, lithium, electric battery, gigafactory
8	sanitation	waste, water, treatment, wasteland, sorting, plant, recycling, treatment plant, collection, center
9	automotive	renault, toyota, car, vehicle, dealership, electric, transport
10	biscuit factory	coffee, production, biscuit, brand, factory, pastry, catering, product, organic
11	cafeteria	central kitchen, kitchen, meal, central, produce meal, school, municipality
12	gas trade	hydrogen, energy, mcphy, hydrogen production, station, production, renewable, fuel, fuel cell
13	urban community	housing unit, housing, town planning, unit, town, residence, planning, city
14	business advice	fusacq, selling business, buying selling, customize, buying, selling, continue accepting, personalize
15	disinfection	mask, coronavirus, production, hydroalcoholic, gel, covid, manufacture, hydroalcoholic gel, health
16	industrial economy	production, factory, investment, turnover, invests, group, employees
17	education and training	school, college, kindergarten, student, building, room, work, city, municipal
18	adult training	recruitment, job, people, recruit, recruiting, contract, group, training, employee
19	rocket	nuclear, plant, nuclear power, power plant, power, nuclear safety, safety, reactor
20	engineering	entreprises group, construction, entreprises, works division, division, special works, works, concrete
21	logistics area	logistics, warehouse, supply chain, supply, warehouses, rhenus, pallet, logistics platform, rhenus logistics

TF-IDF vectorizer measure the importance of words in a collection of documents. This is achieved by multiplying two metrices which are; how many times a word appears in a document, and the inverse document frequency of the word across text documents [2].

For this study, the taxonomy before and after semantic enrichment, and the corpus of 5,000 news articles are converted into a matrix of word vectors through the CV and TF-IDF vectorizer. These word vectors were later compared using the cosine similarity technique to find the most relevant taxonomy concepts for each news article. For instance; the taxonomy concepts which were produced from the CV/TF-IDF vectorizer for a news article are: {Construction of Premises, Moving Transport and Logistics}. Whereas, the manually-labeled taxonomy concepts for this news article are: {Exceptional Transport, Rail Transport, River Transport, Air Transport, Transport and logistics}. The tagged concepts are semantically correlated, which means that even the exact taxonomy concept is not tagged with the text, but there is a possibility that another concept that is a child or parent or another semantically-related concept is labeled with the text by the model. This type of labeling cannot be treated as a completely true or completely false classification. For this instance, distinct true positive and true negative results cannot be obtained for the calculation of classifier accuracy, precision, recall, and the F1 score metrics. To compare the accuracy of this type of classification where the tagged concepts can be semantically correlated, we need to compute the similarity scores of the above-mentioned sets of taxonomy concepts. Words can be similar in two ways, which are; lexically (i.e. have a similar character sequence) or semantically (i.e. have the same context). Word similarity is a value between 0 and 1 which states how close two words are. Different approaches exist in the literature [23], such as Hamming distance, Levenshtein distance, Cosine similarity, etc. to compute text similarity lexically as well as semantically. For our case, to evaluate the classification results, the Cosine similarity measure is used to find similarity between word vectors in the vector space retrieved from spaCy's language model i.e. en-core-web-lg (a collection of over 1 million-word vector). Cosine similarity is measured by the cosine of the angle between two vectors and determines whether the two vectors are pointing in roughly the same direction [23]. The similarity scores among the concepts are computed and later the mean of these scores is calculated to obtain an overall similarity value. Table 3 displays the classification results obtained using the old taxonomy and the new taxonomy. In our case, the TF-IDF vectorizer performed better than the CV.

TABLE III. OVERALL CLASSIFICATION ACCURACY

No.	Taxonomy Classification	Count Vectorizer	TF-IDF Vectorizer
1	Without Enrichment	35%	43%
2	With Topic Enrichment	89%	92%

V. DISCUSSION AND CONCLUSION

New ideas emerge over time in dynamic business environments that create the process of classifying the news challenging due to the flow of diverse information. Moreover, the stride of diverse business news and its updates regarding the products or services have made it more complex [24, 25]. The concepts concerning products or services (e.g. trademarks) in a business market evolve. For instance, certain trademarks (known as genericized trademarks) become generic terms (i.e. a distinctive identifier) of products and services because of their popularity or importance in the market without the intentions of the trademark's owner [26]. For example, Kleenex is a trademark registered with the U.S. Patent Office [27]. It is now used widely by many people in France to describe any facial tissue. Also, Sopalin is a French trademark and is used commonly to call any paper towel. Moreover, Tupperware is the French trademark and is used often to name any sort of plastic box to store food [27]. There are plenty more examples exist in the business market of such trademarks. We need to add them to the company taxonomy such as Kleenex to 'Tissue paper', Sopalin to 'Paper Towel', and Tupperware to 'Plastic') accordingly by linking them with the existing business concepts to incorporate the market evolution for improved business news classification. Hence, business taxonomies should be adjusted (i.e. adding more concepts and removing the old ones) to automate the classifying process of information to reduce human efforts by increasing their productivity and time. In the literature, considerable interest is drawn to the extraction of hypernyms and taxonomic relations from textual data (i.e. lexical datasets such as WordNet) to enrich taxonomies. However, the automatic semantic taxonomy enrichment methods are still not widely developed and applied. Moreover, the existing literature has a shortage of comprehensive studies which attempt to classify the information using up-to-date taxonomy by capturing new market concepts in this specific domain of the business industry.

To cover this gap, this article presented the application of a contextualized topic model (i.e. BERTopic), which has not been explored previously to the best of our knowledge, to automatically enrich the business taxonomy by creating lists of clarifying keywords for each concept in a given taxonomy. BERT-based embeddings built on contextual representations for topic modeling have been proved to be a simple, powerful, and effective method for taxonomy enrichment. The series of contexts (as topics) acquired using this technique in this study are sufficient to add new concepts to the taxonomy. The new concepts are obtained by leveraging a corpus of online news documents which were retrieved from their official websites. The news documents initially were in French and English. The French documents were translated into English to generate model visualizations for this article to show a proof-of-concept application. However, there is a loss of contextual information which can be produced by the document translation. BERTopic allows us to avoid this information loss by processing the news articles in their original language using "multilingual" model configurations. The benefit of applying the topic modeling technique to the corpus of the latest news articles is that an entire set of clusters carrying contextually-similar topics (identified emerging business concepts that arise in the market over time) are mapped with the taxonomy concepts based on their similarity index. The classification results show that augmenting topics from the text corpus into the taxonomy is beneficial for improved taxonomy interpretation and substantially increases the classification accuracy of business documents.

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